

LLVM 21 – Full Changes & Feature Usage Guide

This guide covers **all major changes in LLVM 21**, including removed features, new features, and usage examples for migrating your code.

1. Opaque Pointers

Changes

- Pointer types no longer store element type.
- Removed methods: `getElementType()`, `getPointerElementType()`, `getNonOpaquePointerElementType()`.
- `StructType::getName()` still works for named structs.

Usage Examples

```
// Load instruction
LoadInst *LI = ...;
Type *loadedTy = LI->getType();

// Store instruction
StoreInst *SI = ...;
Type *storedTy = SI->getValueOperand()->getType();

// GetElementPtr instruction
GetElementPtrInst *GEP = ...;
Type *srcElemTy = GEP->getSourceElementType();

// Alloca instruction
AllocaInst *AI = ...;
Type *allocTy = AI->getAllocatedType();

// Global variable
GlobalVariable *GV = ...;
Type *valueTy = GV->getValueType();

// StructType name
StructType *ST = ...;
if (ST->hasName())
    StringRef name = ST->getName();
```

2. Backend Changes

AArch64

- New CPU support: Cortex-A320.
- Scheduling improvements.

AMDGPU

- FWD_PROGRESS bit enabled for GFX ISA >=10.
- New intrinsic: `amdgcnl.load.to.lds`.

LoongArch

- Default code model changes.
- Inline asm and atomic ops support.

Usage Example (C++ Target Query)

```
Triple T(TargetTriple);  
if (T.isAArch64()) {  
    // Apply CPU-specific optimizations  
}
```

3. C API Changes

Removed / New

- Removed some `LLVMConst*` APIs.
- Added `LLVMConstDataArray`, `LLVMGetRawDataValues`.
- New: `LLVMGetICmpSameSign` / `LLVMSetICmpSameSign` for icmp.

Usage Example (C API)

```
LLVMValueRef V = ...;  
LLVMTypeRef T = LLVMTypeOf(V);  
// For GEP  
LLVMValueRef gep = ...;  
LLVMTypeRef elemTy = LLVMGetElementType(LLVMGetOperand(gep, 0));
```

4. Tooling & Debug

- `llvm-objcopy` supports `--update-section`.
- `llvm-strip` continues on errors.
- LLDB now requires Python 3.8+, adds AArch64 watchpoints, improved stepping.

Example LLDB Command

```
(lldb) target create ./a.out
(lldb) break set --name main
(lldb) run
```

5. Libc++ Changes

- `std::is_pod` / `std::is_pod_v` deprecated.
- Removed `packaged_task::result_type`.
- `std::vector<bool>` formatting/performance improvements.

Example Usage

```
#include <type_traits>
static_assert(std::is_standard_layout<int>::value, "Check standard layout");
```

6. Clang Frontend

- ARC migrator removed.
- Pointer arithmetic optimizations; `!nonnull` / `!align` metadata.
- Removed type-trait builtins, deprecated flags.

Example Usage

```
int *p = nullptr;
// Compiler may optimize pointer arithmetic
int *q = p + 1;
```

7. Performance / Pass Infrastructure

- ThinLTO distributed compilation (DTLTO).
- Optimizations for pass scheduling.

Example: Running ThinLTO

```
clang -O3 -flto -fthinlto-index=<index_file> main.c -o main
```

8. Migration Tips

1. Stop using pointer element queries; use instruction-based queries.
 2. Preserve struct names via debug info if needed.
 3. Use metadata to track type info across opaque pointers.
 4. Update C API usage to new methods.
 5. Review removed flags in Clang and libc++.
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9. Useful Links

- LLVM 21.1.0 Release Notes: <https://releases.llvm.org/21.1.0/docs/ReleaseNotes.html>
- Opaque Pointers Guide: <https://llvm.org/docs/OpaquePointers.html>
- LLVM Language Reference: <https://llvm.org/docs/LangRef.html>
- Clang Release Notes: <https://releases.llvm.org/21.1.0/tools/clang/docs/ReleaseNotes.html>
- libc++ Release Notes: <https://fossies.org/linux/llvm-project-llvmorg/libcxx/docs/ReleaseNotes/21.rst>